TENNESSEE STATE UNIVERSITY

Designing an assistive robot for waste Management



Capstone Design Project

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The Team

The Waste Management Challenge

- Over 2 billion metric tons of unsustainable generated waste are thrown away globally every year, entering our environment and impacting climate change, wildlife, and human health (www.earthday.org).

- The total waste generation for 2050 is projected to reach 3.78 billion metric tons if urgent action is not taken.

- One consequence is revealed in a 2021 research that claims 19 million tons of plastic waste enter aquatic environments annually, which is also being consumed by marine life.

- The main goal of this capstone project is to provide a solution for waste management by designing and developing a mobile robot that can identify, collect and sort waste into categories of paper, metal, and plastic.

Project Objectives

- To Successfully design and build an autonomous robot.
- To build the pick and place robot
- To program the robot using python.
- To test the multiple functions of the pick and place robot.



Design Requirements: Functional

- The system shall navigate its environment while avoiding obstacles.
- System will classify waste correctly (Plastic, Metal, & paper)
- The system will transport waste to it’s assigned bins.
- Safely navigating the surrounding environment.
- The process of detecting, moving, and placing said recyclable.

Design Requirements: Non-Functional

- The system shall detect waste at a success rate of 80% or higher. (Prediction before testing)
- The system shall be lightweight.
- The system shall contain user-friendly controls.
- The system shall safety navigate without causing harm to itself or objects.
- The system shall operate for a prolonged amount of time.

Design Limitations

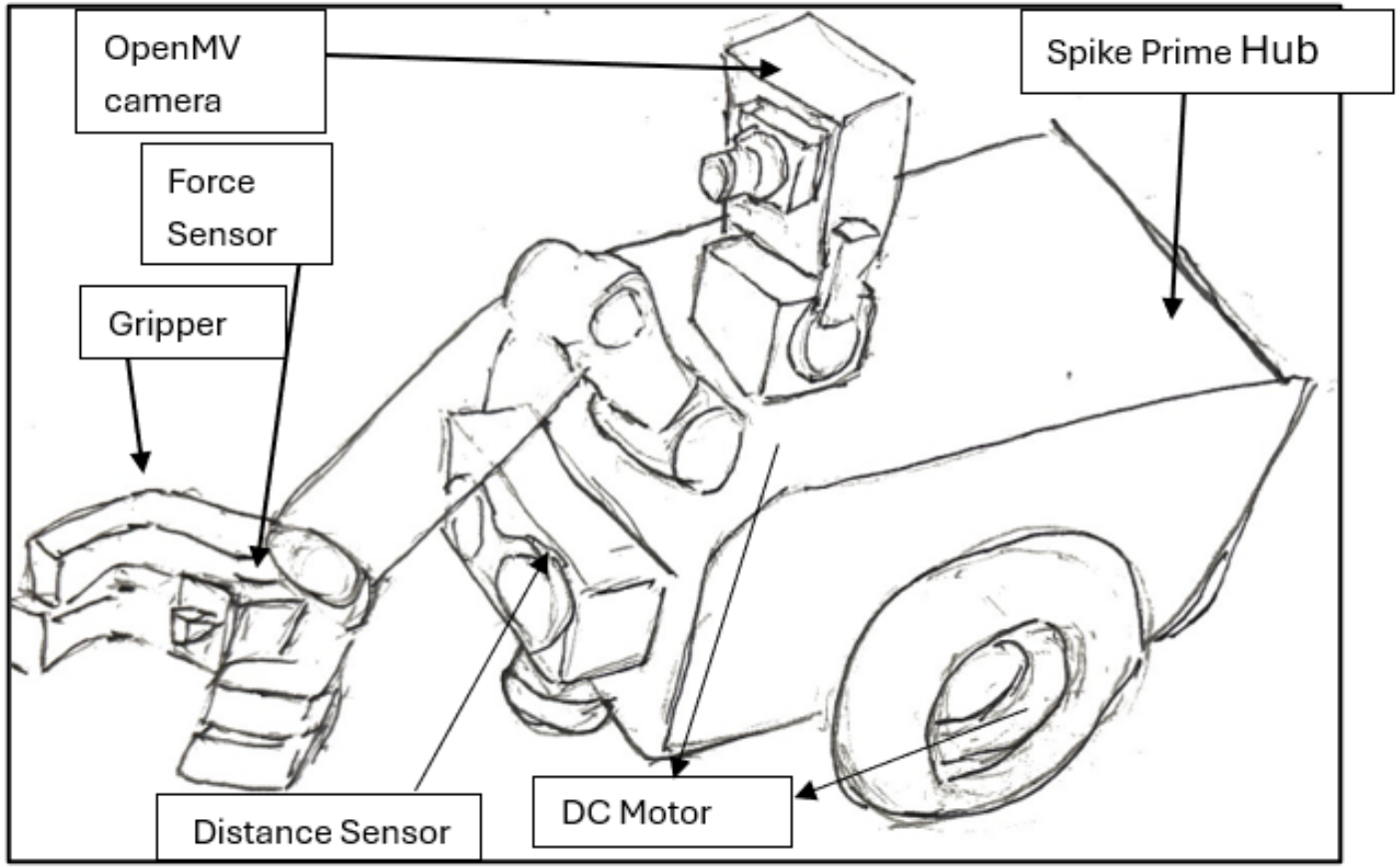
- **Weight:** The Lego Spike equipment only allows picking and placing objects that weigh less than 100 grams.

- **Power:** with the use of a 2100mAh rechargeable battery, the power consumption of the motors and the added microcontrollers reduce operating time. This requires recharging the system from time to time.

- **Size**: With the equipment used and the small scale of the robot, the grippers can only transport objects smaller sizes ranging from 2 to inches in Height and Width

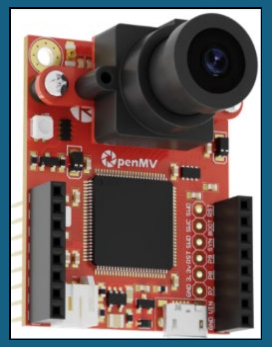
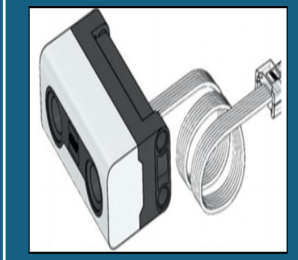
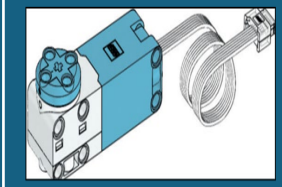
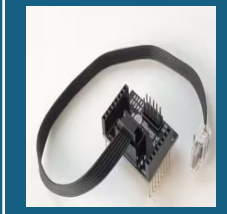
- **Budget:** project was limited by budgetary constraints, which affected the selection of hardware components, sensors, and materials. These eventually restricted extensive prototyping and large-scale testing.

Conceptual Design of Waste Robot

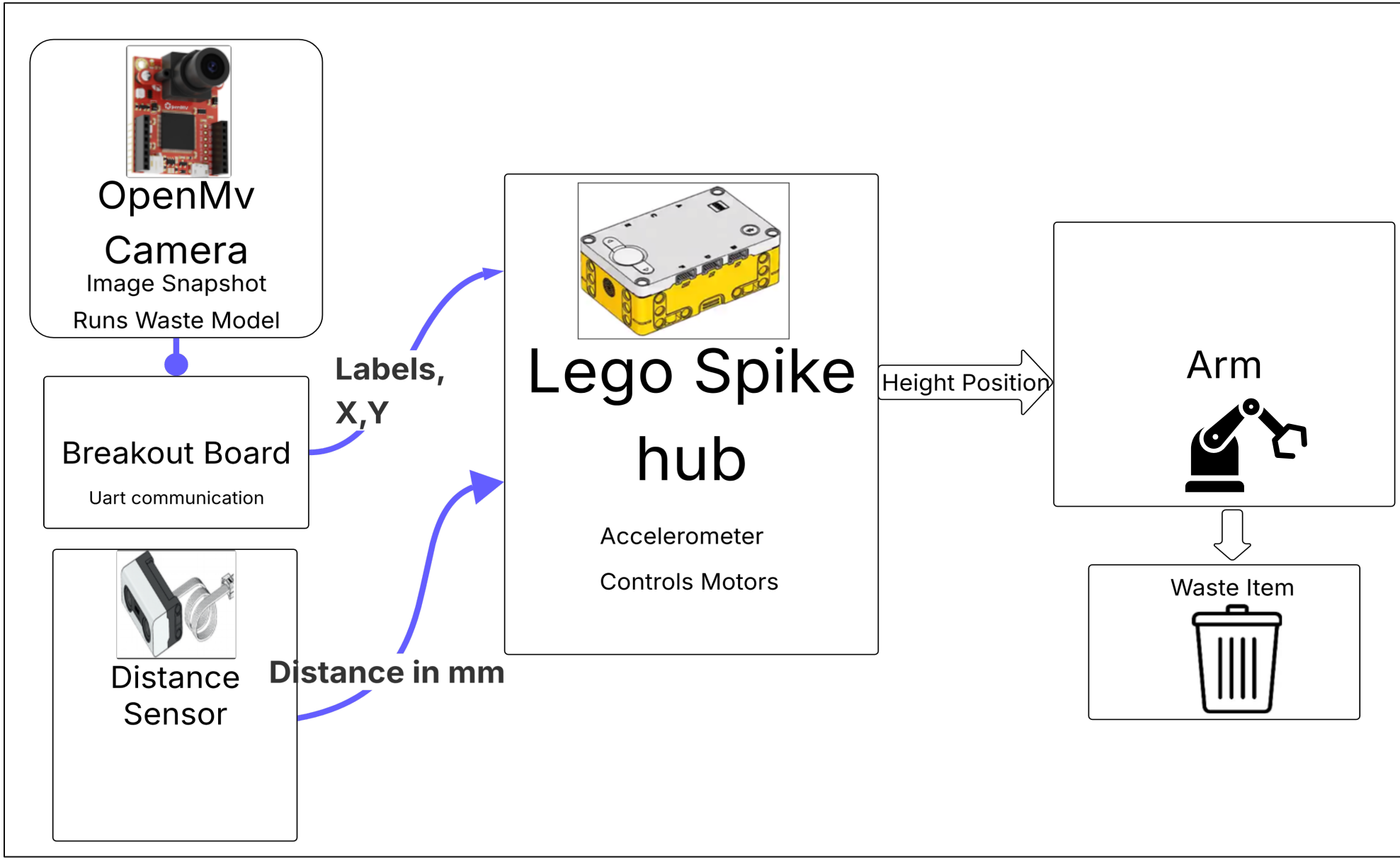


- Arm Length: 12in (Extended)
- Base width: 9in
- Camera distance from the ground: 4in
- Distance sensor from the ground: 1 1/2in
- Height: 1ft 1/2(With arm extended
- Width (Arm included): 14in

Hardware Components

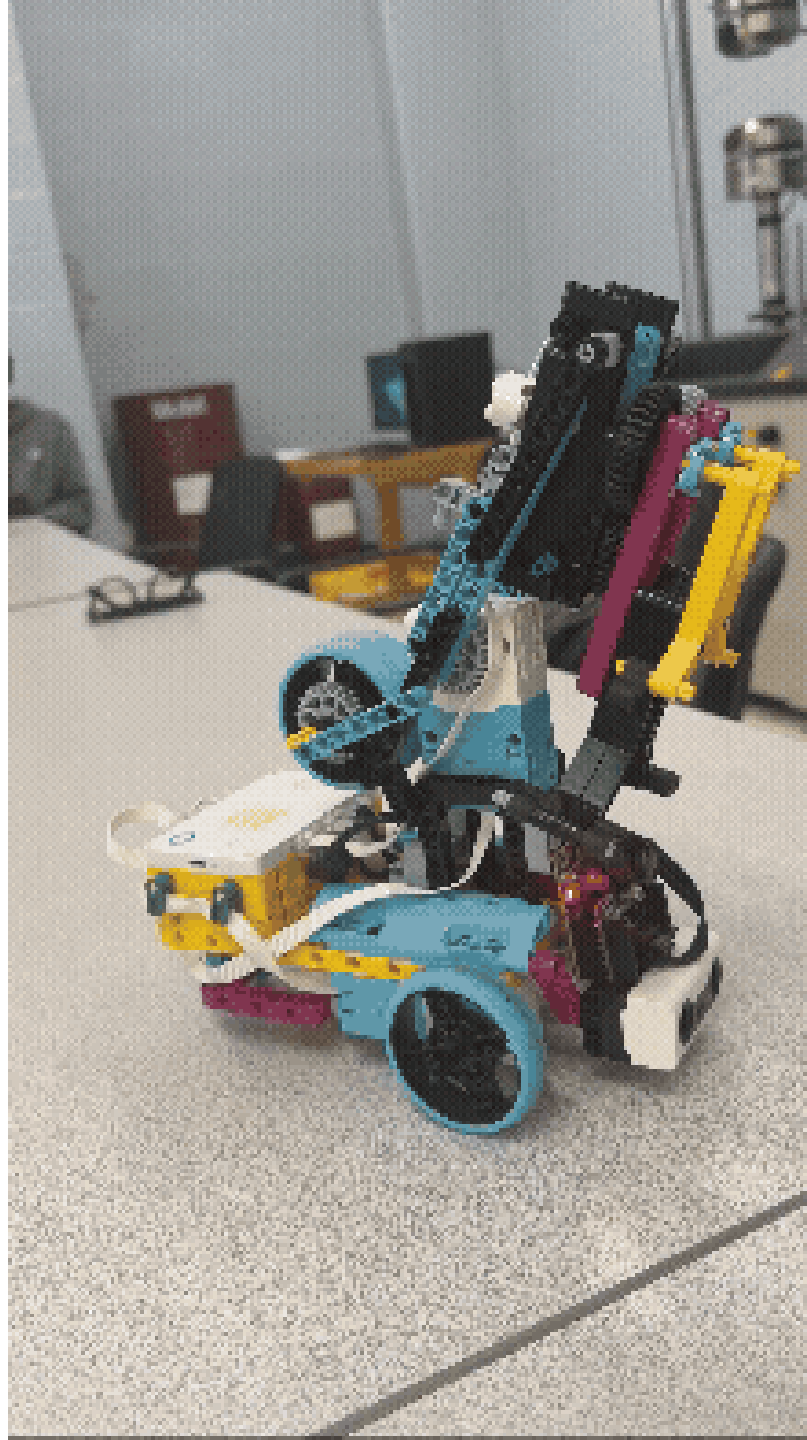
	OpenMV Camera	Lego Spike Prime Kit	Distance Sensor (Lego Spike	DC Motors (Lego Spike Prime)	Antons Spike-OpenMV Breakout Board
					
Function	The camera detects the objects & waste in the surrounding space.	The “foundation” of what both the base and arm are made up of.	Detects distance from objects & waste.	Receiving power from the hub to roll around and lift the gripper.	This board enables communication between third party electronics and the Lego Spike hub
Cost	$60	$0	$0	$0	$32 **Total =$92**

Robot System Architecture

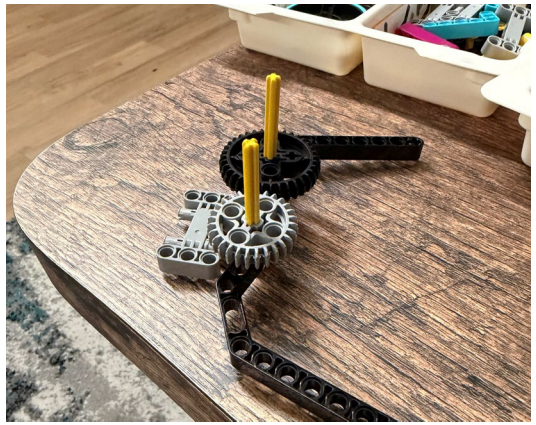


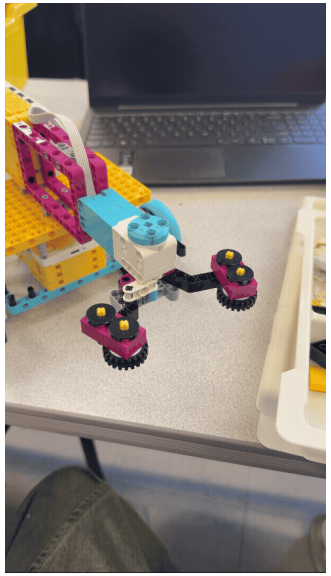
Final Design of Waste Robot

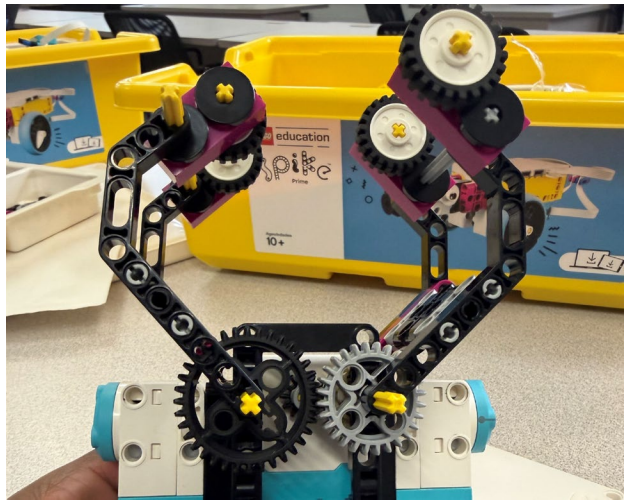
- Arm Length: 10in (Extended)
- Base width: 7 1/2in
- Camera distance from the ground: 3 1/2in
- Distance senor from the ground: 1 3/8in
- Height: 1ft ¼ (With arm extended
- Width (Arm included): 13in

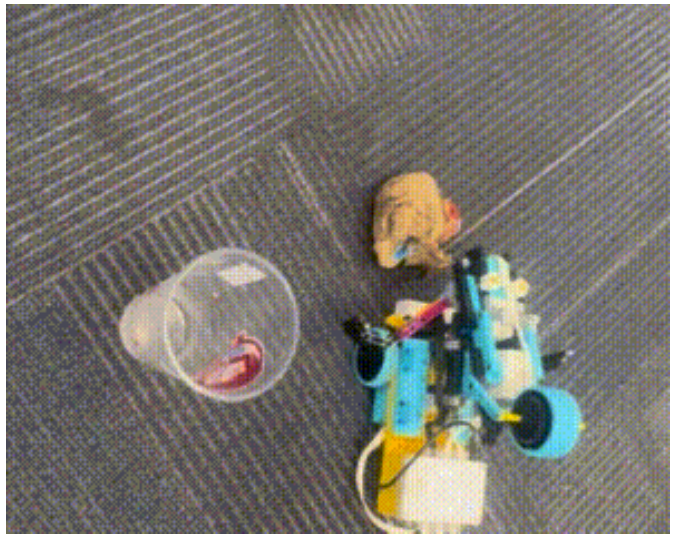


Motorized Gripper Attachment



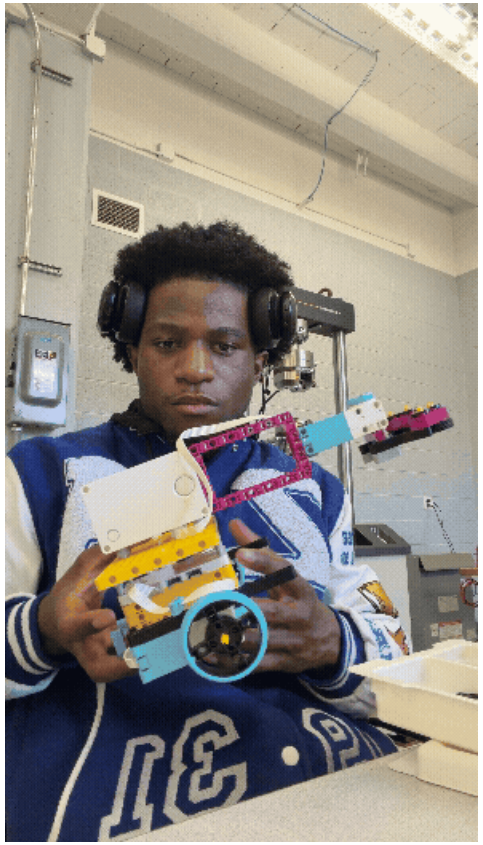


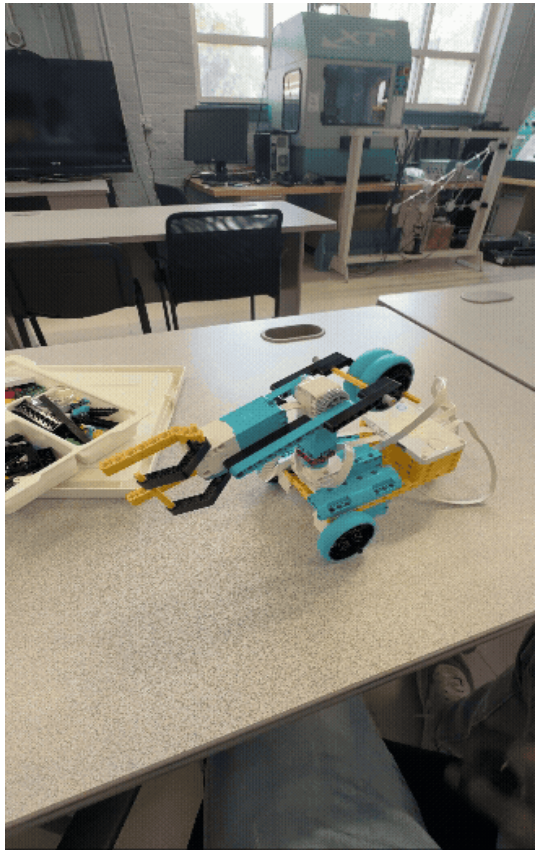


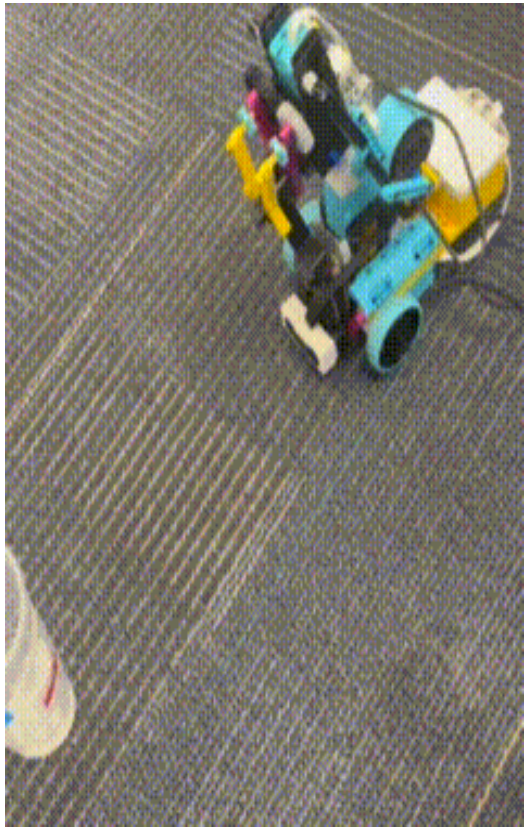


- During this process, multiple iterations of the arm have been designed, created and tested.
- Originally, what started off as a 2-point axis arm eventually became a single piece.

Gripper Arm



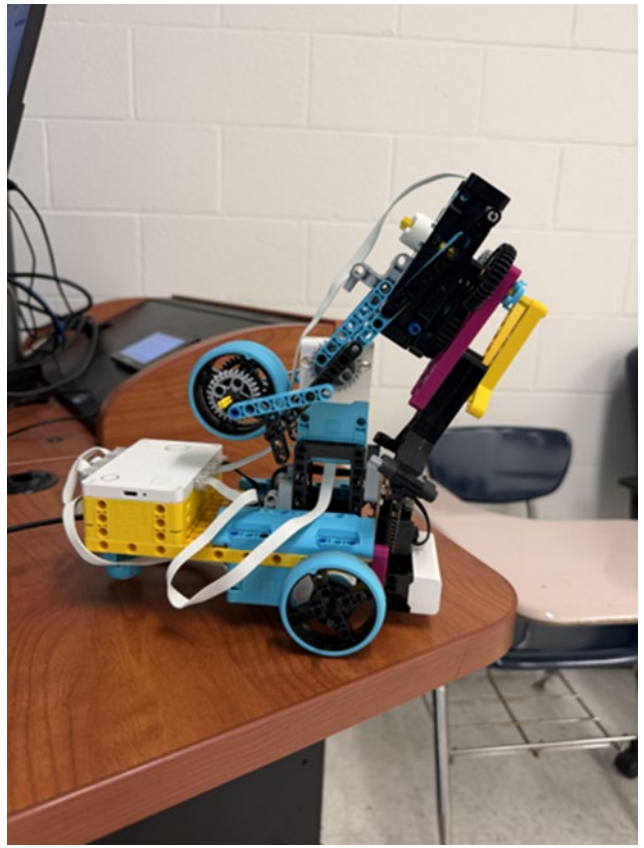




- The arm has had 3 different variations of itself.
- The rubber attachments on the final design are key to gripping

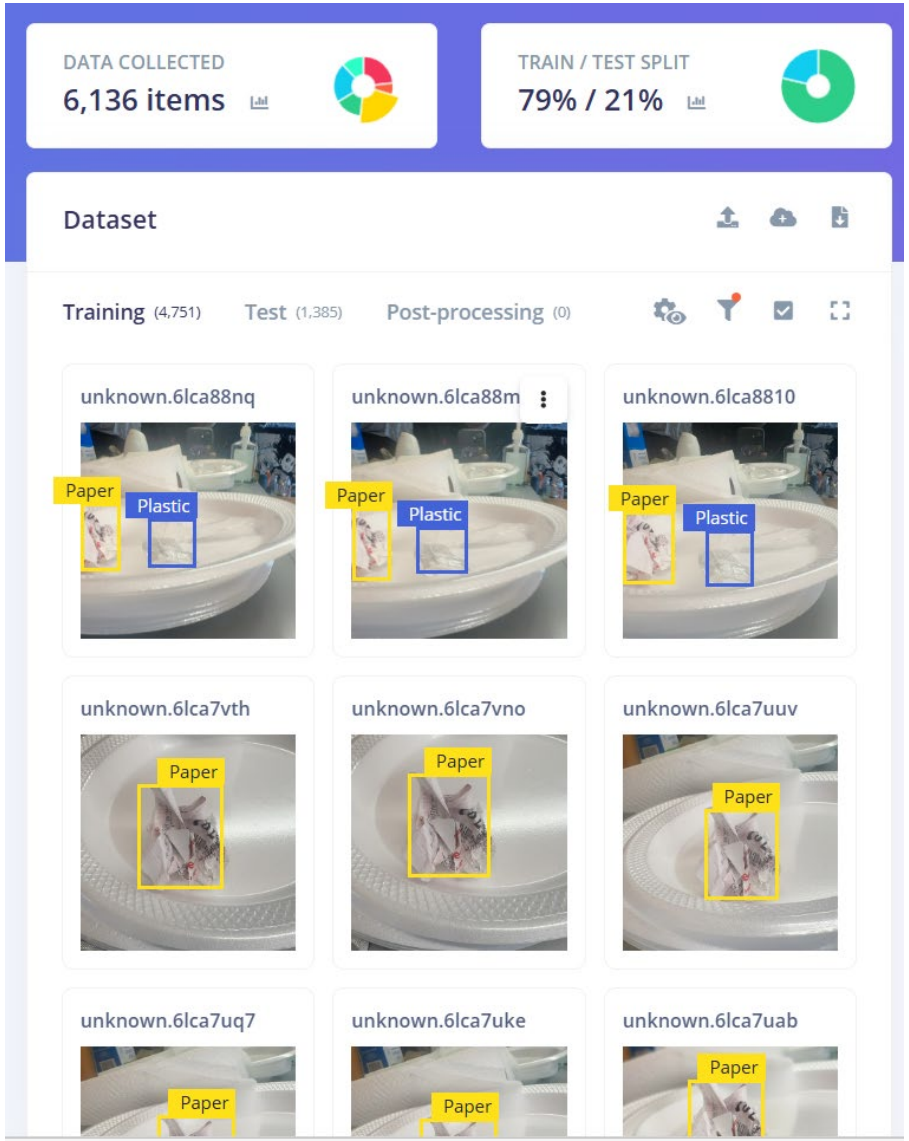
Driving Base

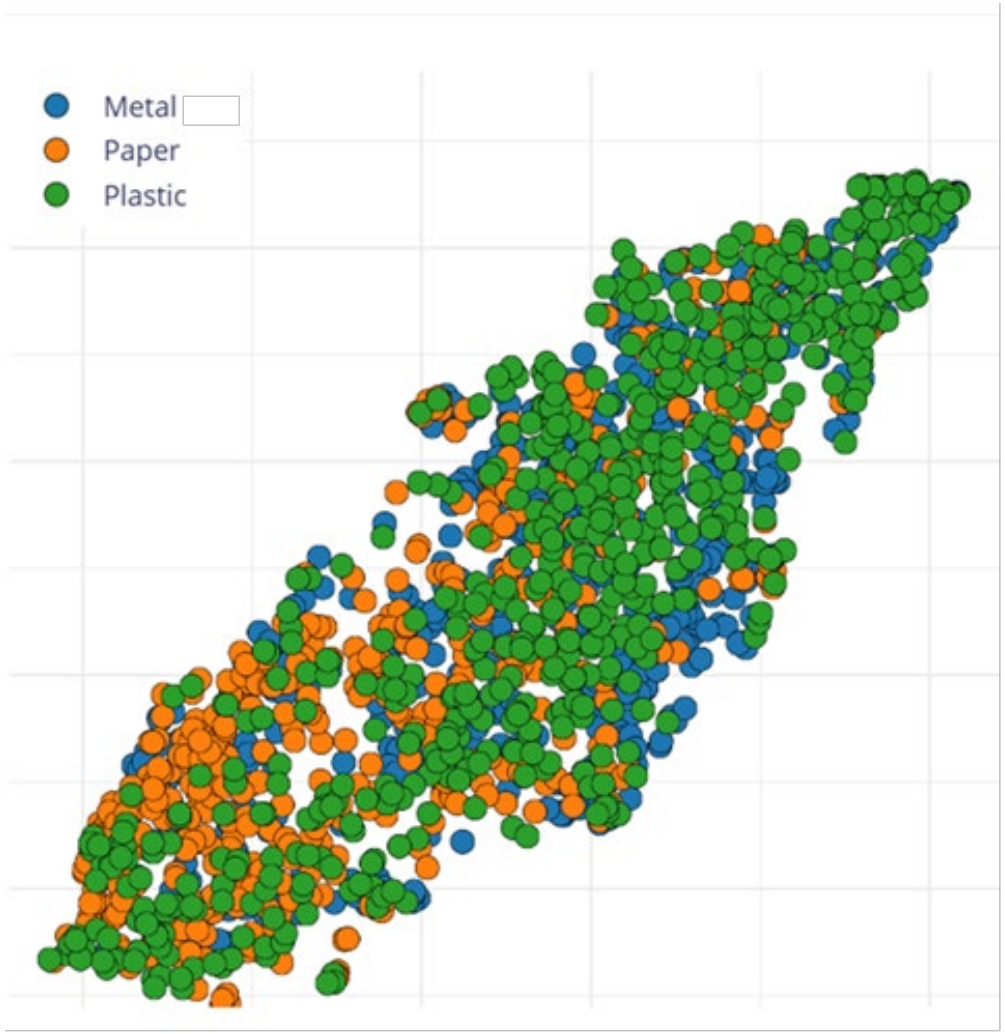




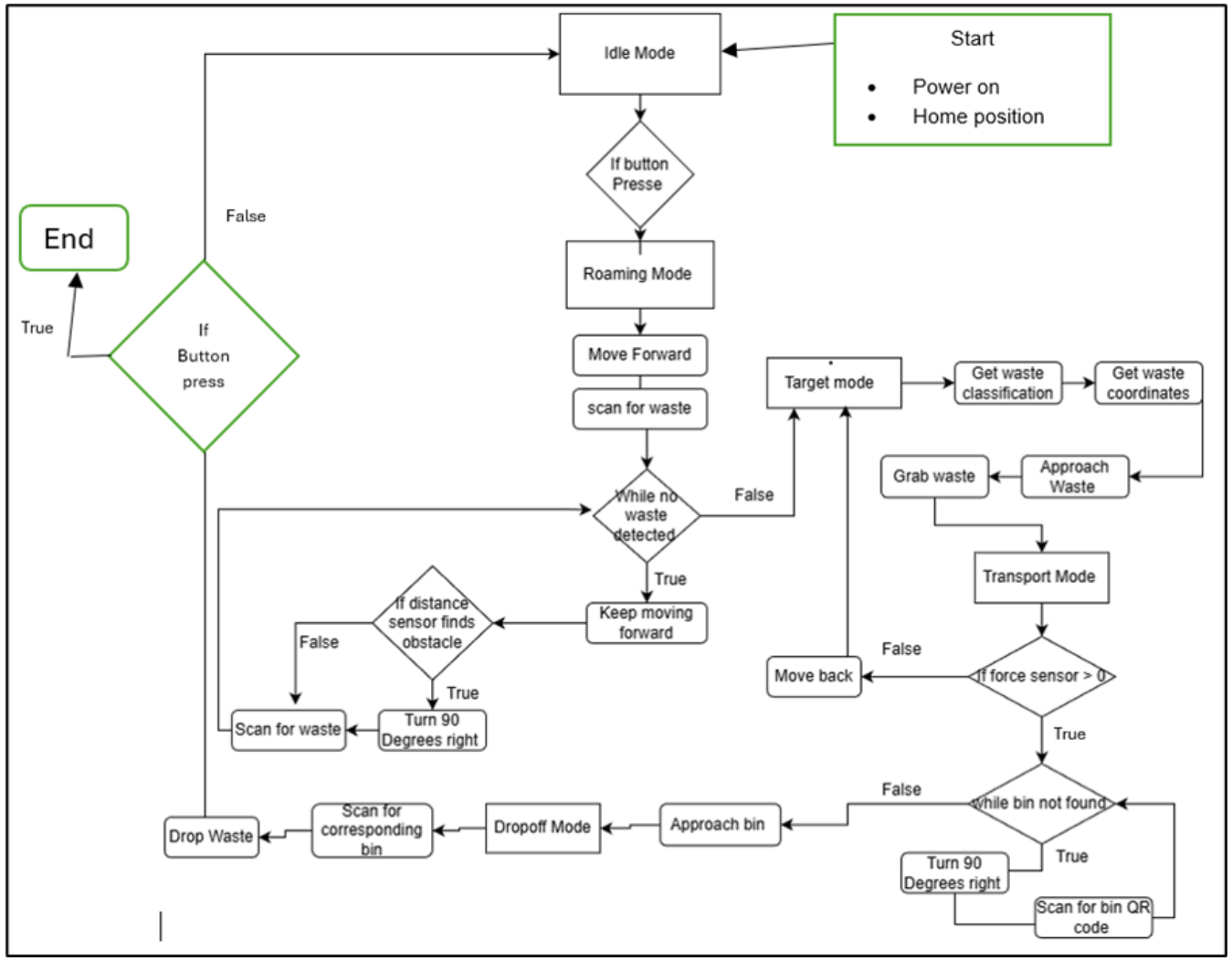
- Only two versions of the base were designed and created.
- First designed included the Hub being underneath the base
- After trial an error, we made the choice to readjust and design a more dynamic bot

Model Training





Robot Programming



Idle Mode – System remains idle until activated

Roaming Mode – System keeps moving whilst dodging obstacle until waste is detected

Target Mode- System approaches, classifies, and grabs waste

Transport mode - System scans for and transports nearby bins

Dropoff Mode – System identifies corresponding bin and releases said waste into bin

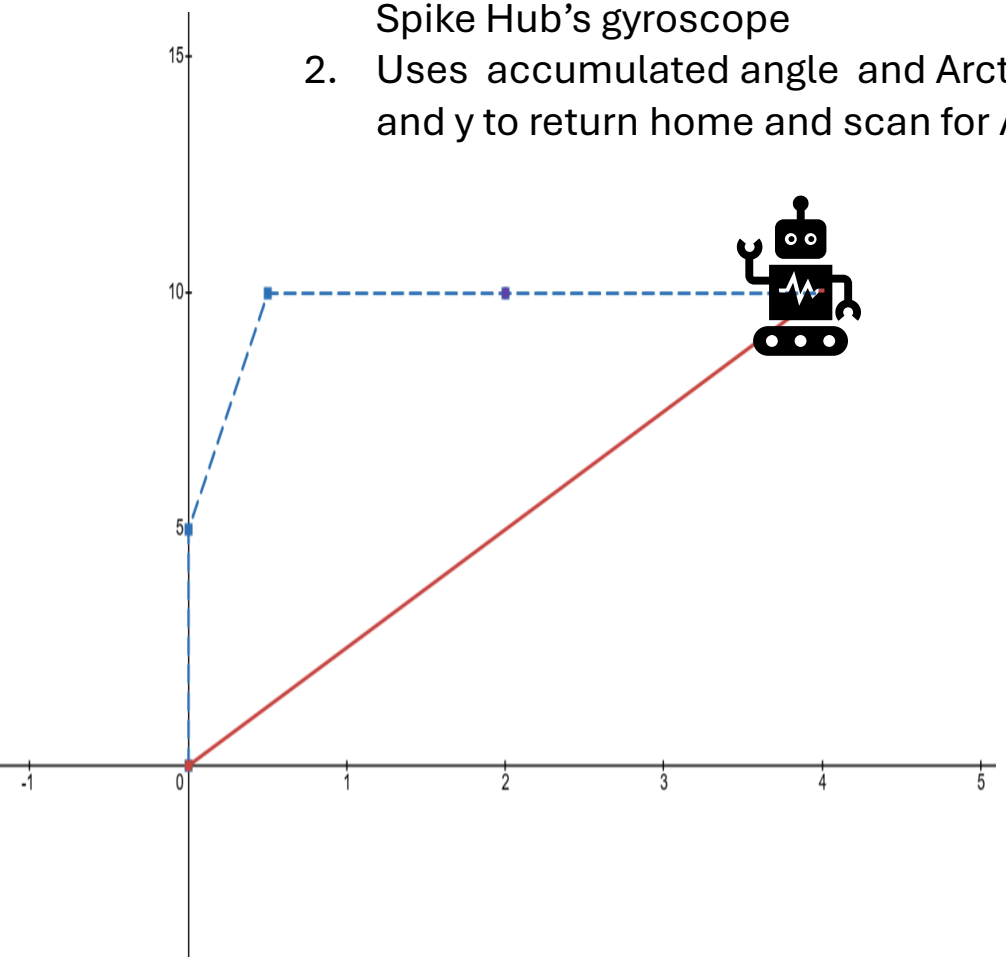
Navigation Method

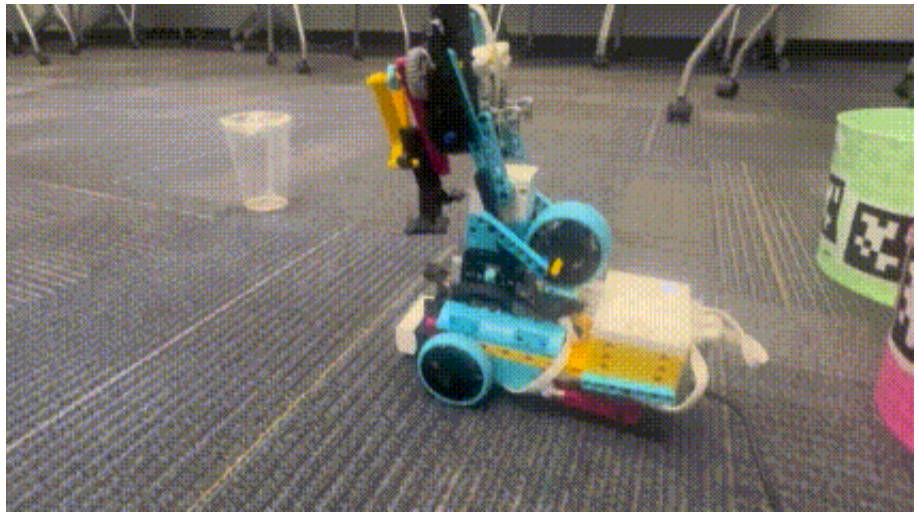
Roaming Mode

- Random turnings and until target found

Dropoff Mode

1. Records x, y, and angle movement from the Spike Hub's gyroscope
2. Uses accumulated angle and Arctangent of x and y to return home and scan for April Tags.







Testing methodology

1. Define Testing: robot accurately identifying and classifying the waste.
2. Set up test procedure: define waste classes Paper , Metal & Plastic.
3. Collect Data: Prepare a test set of waste items| 30 tests per waste class.
4. Run controlled experiment: Robot roams, detects, classifies, and places into bin
   - Record results in the table

Test #	Actual Waste Class	Robot Out Class

5. Build a Confusion matrix.
   - Is a performance evaluation tool used in machine learning and classification problems to show how well a model's predictions match the actual (true) outcomes
6. Calculate Key Performance Metrics

Performance Criteria

Accuracy	Measures success rate	$\frac{TP + TN}{TP + TN + FP + FN}$
Precision	A measure of corrects positive prediction	$\frac{TP}{TP + FP}$
Recall	A measure of actual positives found	$\frac{TN}{TP + FN}$
F1 Score	Measures how well model balances classes	$2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$
Specificity	A measure of negatives found	$\frac{TP}{TN + FP}$

TP	TRUE POSITIVE
TN	TRUE NEGATIVE
FP	FALSE POSITIVE
FN	FALSE NEGATIVE

Confusion Matrix

		Predicted		
		Paper	Metal	Plastic
Actual	Paper	18	0	2
	Metal	0	12	8
	Plastic	0	1	19

Testing Results

Performance Metric	Paper	Metal	Plastic
Precision	1.00	0.923	0.655
Recall	0.900	0.600	0.950
F1-Score	0.947	0.727	0.775
Overall Accuracy			0.817

Conclusions

1. We've successfully designed and built an autonomous robot capable of roaming, detecting, and classifying waste in a controlled environment
2. The robotic system has demonstrated the ability to grasp and drop waste of small sizes into designated bins.
3. Controlled experiments of the system's waste classification model revealed an accuracy of 81%
4. Test results reveal a low accuracy for metal detection during real time trials, due to a bias towards plastic detection
5. Constraints in time and budget hindered model classification accuracy due to missing an internet-based prediction style.

Lessons Learned

1. Integrating an embedded system requires an understanding of communication protocols and pinouts
2. The importance of optimizing for a smaller detection model size on limited hardware without having to compromise for accuracy
3. Latency becomes a problem when trying to introduce a cloud-based detection model to real time robotics system
4. Machine learning model performs better when trained on less broad and more distinct subclasses
5. Don’t overthink! The designing/prototype isn’t meant to be perfect.

Questions?